Part 1 – Iperfer on Wired Environment

Take a screenshot of the measurements (both client and server) when running the tool on the terminal and insert them in a file called answers.pdf under the heading Part 1 - Iperfer on Wired Environment.

```
gklassy@DESKTOP-Q7DADNV CS_640_Partner $ # WIRED, CLIENT
gklassy@DESKTOP-Q7DADNV CS_640_Partner $ java Iperfer -c -h 192.168.0.114 -p 2000 -t 30
sent=2056713 KB rate=548.457 Mbps
```

```
Wills-MacBook-Air:CS_640_Partner Will$ # WIRED, SERVER
Wills-MacBook-Air:CS_640_Partner Will$ java Iperfer -s -p 2000
received=2056713 KB rate=548.183 Mbps
Wills-MacBook-Air:CS_640_Partner Will$
```

Part 1 – Iperfer on Wireless Environment

Under which environment - wired or wireless, would you expect greater throughput?

We would expect a wired environment to have greater throughput. This is because wired connections are, in general, more stable and less subject to interference compared to wireless. As such, wired connections will spend less time retransmitting lost or corrupted packets.

Take a screenshot of the measurements (both client and server) when running the tool on the wireless environment and insert them into answers.pdf.

Select Bash on Ubuntu on Windows

```
gklassy@DESKTOP-Q7DADNV CS_640_Partner $ # WIRELESS, CLIENT
gklassy@DESKTOP-Q7DADNV CS_640_Partner $ java Iperfer -c -h 192.168.0.110 -p 2000 -t 30
sent=62297 KB rate=16.613 Mbps
```

CS_640_Partner — -bash — 80×24

```
Wills-MacBook-Air:CS_640_Partner Will$ # WIRELESS, SERVER
Wills-MacBook-Air:CS_640_Partner Will$ java Iperfer -s -p 2000
received=62297 KB rate=16.592 Mbps
```

Now compare the throughputs between the wired and wireless environments you had obtained. Did it match the prediction you made at the start of this section? In either case, explain your results. Put your answers into answers.pdf

The results matched up with what we expected. The wireless throughput was an order of magnitude worse than the wired throughput. We expected the wireless throughput to be worse, but it was actually much worse than we originally thought, especially when comparing the same two hosts on the same network.

Part 3 – Measurements in Mininet

Q2: Path Latency and Throughput

Now, assume h1 wants to communicate with h4. What is the expected latency and throughput of the path between the hosts?

The total latency should equal the sum of the link latencies through the path from h1 to h4. This is because each link adds additional latency and the packets must travel through all of them. The path from h1 to h4 goes through links L1, L2, and L3. Thus, the total latency should be…

L1 (80.169ms) + L2 (20.300ms) + L3(40.277ms) = **140.746ms**

The total throughput should be the minimum throughput through the links on the path from h1 to h4. This is because the throughput (or "speed") is bottlenecked by the slowest link on the path. The path from h1 to h4 goes through links L1, L2, and L3. Thus, the total throughput should be…

Min(L1(21.557 Mbps), L2(49.820 Mbps), L3(41.536 Mbps)) = **21.557 Mbps**

Put the average RTT and measured throughput in your answers.pdf file and explain the results. If your prediction was wrong, explain why.

We found that the total latency from h1 to h4 was **140.336ms**. This matches our expected value of 140.746ms. We found that the total throughput from h1 to h4 was **21.506 Mbps**. This matches our expected value of 21.557 Mbps.

Q3: Effects of Multiplexing

What is the expected latency and throughput when two pairs of hosts are communicating simultaneously? Three pairs?

The expected latency when two pairs of hosts (or three pairs of hosts) are communicating at the same time should be the same as the latency when only one pair is communicating. This is because latency describes the time it takes for the first packet to reach the other side. Because switches share traffic between hosts, latency should not increase by a noticeable amount when more hosts are communicating.

The expected throughput when multiple pairs of hosts are communicating will be lower for each host pair. However, summing the throughputs of each host pair should still produce a similar total throughput. This is because all pairs of hosts share the same maximum throughput through a link.

Put the average RTT and measured throughput for each pair in your answers.pdf file and explain the results. If your prediction was wrong, explain why.

Two connections latency
(h1,h9) = 140.638ms
(h7,h10) = 140.627ms

Three connections latency
(h1,h9) = 140.324ms
(h7,h10) = 140.328ms
(h8,h4) = 140.328ms

This matches our expected value of 140.336ms, the latency when only one pair of hosts in communicating.

Two connections throughput
(h1,h9) = 11.260 Mbps
(h7,h10) = 10.162 Mbps
Total = 21.422 Mbps

Three connections throughput
(h1,h9) =10.574 Mbps
(h7,h10) = 4.459 Mbps
(h8,h4) = 6.270 Mbps
Total = 21.303 Mbps

This matches what we expected. The totals are around the same as when only one pair of hosts is communicating (21.506 Mbps) however the throughputs for each pair are reduced as they share the total throughput through the switch.

Q4: Effects of Latency

Lastly, assume h1 wants to communicate with h4 at the same time h5 wants to communicate with h6. What is the expected latency and throughput for each pair?

The latency between h1 and h4 should be the sum of the link latencies on the path. The path from h1 to h4 is L1, L2, L3, so we expect the value to be…

L1 (80.169ms) + L2 (20.300ms) + L3(40.277ms) = **140.746ms**

The latency between h5 and h6 should be the sum of the link latencies on the path. The path from h5 to h6 is L2, L4, L5, so we expect the value to be…

L2 (20.300ms) + L4(30.376ms) + L5(30.332ms) = **81.008ms**

For throughput, the path from h1 to h4 is L1 -> L2 -> L3. The path from h5 to h6 is L4 -> L2 -> L5. We need to consider that both paths use the shared link L2. From previous measurements, L2 has a throughput 49.820 Mbps that both pairs of hosts must share.

However, from previous measurements, we know that the throughput of L1 is 21.557 Mbps and the throughput of L4 is 30.879 Mbps. This means that both links being used which go into L2 have a combined total throughput which is about the same as the throughput of L2. Therefore, even though both hosts are using L2, we shouldn't expect any bottlenecks on L2 as the maximum throughput on L2 will not be reached by L4 and L1 alone. As such, we will ignore the shared link in our expected calculations as it will not be a bottleneck.

Our expected throughput from h1 to h4 is…
Min(L1(21.557 Mbps), L3(41.536 Mbps)) = **21.557 Mbps**

Our expected throughput from h5 to h6 is…
Min(L4(30.879 Mbps), L5(30.363 Mbps)) = **30.363 Mbps**

Put the average RTT and measured throughput in your answers.pdf file and explain the results. If your prediction was wrong, explain why.

The measured latency from h1 to h4 was **140.596ms**, close to our expected 140.746ms.
The measured latency from h5 to h6 was **80.575ms**, close to our expected 81.008ms.
The measured throughput from h1 to h4 was **21.129 Mbps**, close to our expected 21.557 Mbps.
The measured throughput from h5 to h6 was **30.303 Mbps**, close to our expected 30.363 Mbps.